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Written exam

Closed book, two hours.

The exam tests what the project cannot: whether you understand *why* the methods work. It draws on the **handouts**, including the derivations. If you have only run the notebooks, you will not pass.

Structure

Part	Questions	Marks	What it tests
A — Foundations	8 short answers	24	Definitions, statements, when a method applies
B — Derivations	2 of 3 offered	30	Reproduce and reason about a derivation from the handouts
C — Method selection	2 scenarios	26	Choose an approach for a described problem and justify it
D — Diagnosis	1 case	20	Given results, say what went wrong and how to check

Part D always presents a flawed study. Finding the flaw is a course learning outcome, so it is never worth fewer than a fifth of the marks.

What you are expected to be able to derive

From the handouts, unaided:

Lesson	Derivation
3	Normal equation from the least-squares objective; the gradient descent update; Ridge as constrained optimisation

Lesson	Derivation
4	The logistic model's cross-entropy from maximum likelihood; why squared error is a poor choice for classification
5	The bias-variance decomposition of expected prediction error
6	The SVM margin as an optimisation problem; what the kernel trick avoids computing
7	Entropy and information gain for a split; boosting as gradient descent in function space
8	PCA as an eigenvalue problem; the k-means objective and why the algorithm converges
9	Backpropagation for a two-layer network; why non-linear activations are necessary

You are not asked to memorise library APIs. You may be asked to write a few lines of pseudocode.

Materials

- [question_bank.md](#) — questions by topic, built up as the course runs
 - [sample_paper_1.md](#), [sample_paper_2.md](#) — full papers with solutions, published after Lesson 8
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How to prepare

Work through the derivations with a pen rather than reading them. Then take the quiz for each lesson: the quizzes deliberately include questions that require reasoning about a derivation rather than recalling a fact, and those are the closest thing to Part B you will find before the sample papers appear.